

Unit lattices and finite Weil dimensions

Complete power-order unit bases and equation ranks through level 128

Finite residue images recover exact cubic unit lattices. Nonmonic source constraints survive maximal-order coordinates, while original Fourier/chirp equations cross-check the dimension formula.

3

complete unit bases

6

nonmonic source lattices

128

maximum equation level

Certified external maximal-order generators, exact finite-index conversion and source arithmetic. General norm representatives, effective Skolem bounds and Lean refinement remain open.

Source, corpus receipts and reproduction commands accompany the repository snapshot.

This continuation supplies complete unit bases for three cubic power orders that the earlier bounded search left unpriced. The orders are $R = \mathbb{Z}[\theta]$, with $\theta^3 = P\theta + Q$ at $(P, Q) = (39, 2)$, $(48, 30)$ and $(195, 830)$. Certified maximal-order generators are converted into exact generators of R 's unit group through a finite residue image. Six linked nonmonic binary-cubic sources acquire exact coordinate lattices and signed unit-orbit extraction.

A separate calculation closes the requested cross-check of the Weil commutant formula beyond level 64. Ten selected levels, including odd prime powers, powers of two and mixed composites, are checked through their original Fourier/chirp equations up to level 128. Their dimensions agree with the all-level paper formula. These finite computations strengthen the evidence for that formula without replacing its paper proof or claiming a new Lean theorem.

The difference between an order and its unit group

An integral power basis can be a proper sublattice of the maximal order O . For the three fields, the additive indices $[O:R]$ are 6, 7 and 64. Their field discriminants are 6588, 8532 and 2700. The discriminants of the power polynomials are respectively 237168, 418068 and 11059200, and the ratio is the square of the additive index.

The unit indices are different objects. Here $[O^*:R^*]$ equals 6, 8 and 64. In particular, an additive index of seven produces a unit index of eight. A fundamental unit of O can have fractional power-basis coordinates and therefore fail to lie in R . Reusing maximal-order units directly would lose the source order's arithmetic domain.

PARI supplies integral bases and two fundamental units for each totally real cubic field. Every `bnfcertify` call returns one and every `nfcertify` call returns an empty list. The signatures are (3,0), and the only roots of unity are plus and minus one. Complete maximal-order unit generation is the external certified number-field input. The finite conversion into R is exact integer arithmetic.

A finite image determines the infinite exponent lattice

Let $m = [O:R]$. Since O/R has order m , multiplication by m annihilates it, hence mO is contained in R . Reducing integral-basis coordinates modulo m gives the finite ring O/mO . The subgroup H generated by the residue images of the two fundamental units is traversed until every generator edge stays inside the retained state set.

A state c belongs to the image of R precisely when its power-basis coordinates are integral. If B is the matrix taking integral-basis coordinates to power coordinates, the test is $mBc = 0$ modulo m . The compiler also checks that the power basis embeds integrally into O and that m clears the reverse basis. Thus neither inclusion is merely inferred from a discriminant ratio.

The intersection of H with the image of R is a group. An O -unit lying in R is automatically an R -unit: its integral characteristic polynomial has constant coefficient plus or minus one,

so its inverse is an integer polynomial in the element and also lies in R . Both unit groups contain minus one, so the torsion sign does not change the index calculation.

Each finite state has an exponent label (r,s) . A generator edge from label v to label w supplies a kernel relation $v+e_i-w$. The spanning-tree relations generate the kernel of the map Z^2 to H : an arbitrary closed word decomposes into these edge loops. Adding the labels of all states in the image of R gives exactly the preimage lattice L of allowable unit exponents.

Two-dimensional Hermite reduction writes L with column basis $(a,0),(b,g)$, where a and g are positive and $0 \leq b < a$. Membership is equivalent to g dividing s and a dividing $r-b(s/g)$. Its index is ag . Independently, the finite image gives $[H:H_R]=|H|/|H_R|$. The two indices agree, and every retained state label has the predicted lattice membership.

The three complete unit bases

For $\theta^3=39\theta+2$, H has six states and only one lies in the power-order residue subgroup. The exponent columns are $(6,0),(5,1)$. In terms of maximal-order generators η_1 and η_2 , every power-order unit is a signed product of η_1^6 and $\eta_1^5 \eta_2$. Their power coordinates are $(5253,103138,16448)$ and $(-660981,-12994541,-2072335)$.

For $\theta^3=48\theta+30$, H has 96 states and twelve lie in the power-order subgroup. The exponent columns are $(4,0),(1,2)$, giving unit index eight. The power-coordinate generators are $(24121,41930,5806)$ and $(23267,40449,5601)$. This removes the earlier failure to find generators by a small coordinate search.

For $\theta^3=195\theta+830$, H has 1024 states and sixteen lie in the power-order subgroup. The exponent columns are $(16,0),(14,4)$, giving unit index 64. The first generator has coordinates $(-2779851436213158215669,-398133592302353045509,36516248873178164027)$; the second has $(8638027502305261,243456662042202,-50336068403964)$. Their size explains why a bounded search in power coordinates was ineffective.

Every compiled generator and inverse has integer power coordinates. Their product is exactly one in the original cubic algebra, and each determinant norm is plus or minus one. The maximal-order multiplication matrices and their integer inverses are retained for signed orbit calculations. The complete basis result uses the certified maximal-order generators; it is not a claim that the old Lean unit-enumeration costs have become zero.

Nonmonic sources retain two divisibility conditions

Let $F(r,s)=ar^3+br^2s+crs^2+ds^3$, with nonzero a , and let $\phi=h+k\theta$ with nonzero integer k . The compiler checks the exact polynomial identity $\phi^3+b\phi^2+ac\phi+a^2d=0$. It follows that $\text{Norm}(ar-s\phi)=a^2F(r,s)$. This is the same nonmonic norm transport as before, now expressed in the actual maximal-order basis.

Write the power coordinates of an element as (u,v,w) . The source image consists exactly of $w=0$, k dividing v , and a dividing $u-hv/k$. The reverse map is $s=-v/k$ and $r=(u-hv/k)/a$.

The scale k must be retained alongside the leading coefficient a ; dropping either divisibility condition admits elements outside the original binary source.

For integral-basis coordinates c , put $\text{ell}_i(c) = m$ times its i -th power coordinate. All ell_i have integer coefficients. The source conditions become $\text{ell}_2(c) = 0$, $m k$ dividing $\text{ell}_1(c)$, and $m k a$ dividing $k \text{ell}_0(c) - h \text{ell}_1(c)$. These conditions are exact even when an arbitrary maximal-order element has fractional power coordinates.

The six sources have leading coefficients -7 , -6 and -4 , hence target norm multipliers 49 , 36 and 16 . Their linear generators include $\phi = 15 + 16\theta$, $\phi = 11 + 8\theta$, $\phi = 2 + 4\theta$ and $\phi = 1 + 2\theta$ in the $P = 48$ field, $\phi = 1 + 13\theta$ in the $P = 39$ field, and $\phi = 7 + \theta$ in the $P = 195$ field. The retained source lattices recover every signed source pair exactly.

Signed orbit extraction and its remaining bound

For a representative Γ and the compiled power-order units ϵ_1 and ϵ_2 , multiplication is given by two commuting integer matrices M_1 and M_2 . Every orbit element is $M_1^r M_2^s \Gamma$. The inverse matrices supply negative exponents exactly. Applying ell_2 gives the scalar zero constraint, while the two congruence rows preserve the source divisibilities.

Whenever all three source conditions hold, the reverse map returns the original pair $(r_{\text{source}}, s_{\text{source}})$ and its binary-cubic value. Multiplication by a unit preserves the absolute norm. This supplies the arithmetic operator and extraction stage of a Skolem-style solver without silently substituting an unrestricted norm equation for the source equation.

A complete finite set of norm representatives and a proven bound or zero theorem for the exponent orbit are still needed to solve every source globally. The current runtime evaluates signed exponents of absolute value at most 64 ; the matrix identities themselves hold for every integer exponent. It does not declare unsuccessful finite orbit searches complete.

Original-equation Weil dimensions beyond 64

Use $F[x, y] = \zeta_N^{xy}$ and $T[x, x] = \zeta_N^{x^2}$. Commutation with T forces an operator entry $A[x, y]$ to vanish unless $x^2 = y^2$ modulo N . The surviving entries are unknowns in the equations $AF = FA$. Explicit rational operators provide a lower dimension bound d after their cyclotomic commutators and independence are checked exactly.

For an upper bound, specialize ζ_N to a primitive N -th root in a prime field. A selected original-equation minor has nonzero determinant modulo that prime. Its determinant is therefore nonzero over the cyclotomic field as well. If the number of surviving entries is M and the minor has rank $M - d$, the commutant dimension is at most d . Combining the bounds gives equality.

The resulting dimensions are $c(65)=4$, $c(72)=15$, $c(81)=5$, $c(96)=18$, $c(100)=9$, $c(108)=12$, $c(112)=14$, $c(121)=3$, $c(125)=4$ and $c(128)=13$. They agree with $c(N)=\tau(N)$ for odd N , and $c(2^a m)=(2a-1)\tau(m)$ for odd m and $a \geq 1$. This comparison uses the formula only after constructing and checking the original-equation packets.

At $N=128$ there are 896 allowed entries, thirteen independent rational commuting operators, and a nonzero minor of rank 883 over the field with 257 elements and primitive root nine. At $N=96$ the corresponding counts are 800, eighteen and 782. Both are independent finite rank calculations beyond the former dense limit.

The exact replay now constructs each original equation row once before selecting its minor columns. Previously it reconstructed that row for every selected column. An optional integer-array path accelerates modular elimination; all entries are reduced below 100000, so every product is below 10^{10} and fits signed 64-bit arithmetic. No floating-point rank threshold enters the result. The standard-library path remains available.

Reproduction and scope

Run `make power-order-unit-receipts` with PARI/GP installed or `PERFECTPOWER_GP` set. Run `make weil-beyond64-receipts` to regenerate or replay the ten equation packets, and `make check-units-weil` for the focused tests. Weil packets are stored together in a compressed JSON census; all exact operators and selected equation minors are retained.

The delivered closures are complete computational unit bases for the three named orders, exact nonmonic source lattices and signed extraction, and the requested beyond-64 finite dimension cross-check. General norm representatives, effective Skolem bounds, arbitrary power-order compilation and Lean unit/descent refinement remain separate. The dense spectral and CRT-product interfaces retain their previous level-64 limits; the original-equation dimension interface now supports levels through 128.

Sources

PARI's certified unit and integral-basis algorithms are documented at https://pari.math.u-bordeaux.fr/dochtml/html-stable/General_number_fields.html. The dimension formula and monomial-action proof are in `docs/WEIL_LOCAL_DIMENSION_MONOGRAPH.md`. The original finite rank-minor construction is in `python/perfectpower/weil_commutant.py`; the power-order conversion and source compiler are in `python/perfectpower/power_order_units.py`.